

CHELYABINSK GOROD, Russian Federation

WMO: 286450

Lat: 55.150N

Lon: 61.300E

Elev: 778

StdP: 14.29

Time Zone: 5.00 (E05)

Period: 04-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-17.5	-28.4	1.2	-22.8	-22.0	1.7	-16.8	13.4	5.4	11.3	13.5	1.6	0	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	86.7	66.4	83.5	65.9	80.6	64.6	69.9	81.8	68.4	79.3	66.7	77.0	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	97.8	75.3	64.3	92.7	74.2	62.7	87.7	72.4	34.3	82.1	33.1	79.5	31.7	77.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.3	10.7	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-30.5	91.7	7.8	5.5	-36.1	95.7	-40.7	98.9	-45.0	102.0	-50.7	106.1		
			-30.6	72.4	7.8	1.2	-36.2	73.3	-40.8	74.0	-45.1	74.7	-50.8	75.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	3.7	7.8	22.5	42.0	56.0	64.7	67.0	64.4	53.4	40.9	24.8	11.0	(6)
(7)		DBStd	24.39	11.62	10.83	8.87	9.03	8.48	7.85	7.31	7.18	7.67	8.17	11.95	12.98	(7)
(8)		HDD50	6098	1436	1182	854	269	38	6	0	1	49	300	755	1208	(8)
(9)		HDD65	10059	1901	1602	1319	691	298	98	67	101	356	748	1205	1673	(9)
(10)		CDD50	1841	0	0	0	27	226	447	527	447	150	17	0	0	(10)
(11)		CDD65	326	0	0	0	0	20	88	129	82	7	0	0	0	(11)
(12)		CDH74	2879	0	0	0	13	263	792	1106	648	57	0	0	0	(12)
(13)		CDH80	765	0	0	0	1	41	209	322	185	7	0	0	0	(13)
(14)	Wind	WSAvg	3.5	2.9	3.1	4.2	4.4	3.7	3.4	3.1	2.9	3.1	3.8	3.9	3.2	(14)
(15)	Precipitation	PrecAvg	25.4	1.7	1.3	1.6	1.5	2.1	2.9	3.4	2.8	2.0	2.3	2.1	1.8	(15)
(16)		PrecMax	32.3	4.2	2.5	3.4	3.9	4.5	4.8	7.5	4.6	3.7	5.2	4.0	3.7	(16)
(17)		PrecMin	18.2	0.4	0.3	0.1	0.1	0.6	1.4	1.5	0.9	0.3	0.4	0.2	0.6	(17)
(18)		PrecStd	3.7	0.8	0.6	0.8	0.9	1.0	0.9	1.6	1.0	0.8	1.1	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	38.0	50.1	75.1	84.6	89.5	91.8	90.6	81.2	65.7	48.2	40.1	(19)
(20)		MCWB	32.4	34.6	41.8	56.1	61.3	66.3	67.7	67.0	61.9	52.1	42.9	34.7	(20)	
(21)		2%	DB	29.5	32.9	42.2	67.2	80.6	85.6	86.6	84.7	74.5	61.7	43.8	35.4	(21)
(22)		MCWB	27.2	29.8	37.0	51.2	60.0	65.2	67.3	66.8	59.9	50.3	40.2	32.7	(22)	
(23)		5%	DB	25.3	28.9	38.2	61.3	77.1	81.7	83.5	80.9	70.0	57.6	40.8	32.7	(23)
(24)		MCWB	23.6	25.9	33.9	48.4	59.2	65.1	67.2	66.3	58.4	48.5	37.9	30.7	(24)	
(25)		10%	DB	21.5	24.3	35.3	56.3	72.6	78.6	80.2	76.9	65.6	53.3	38.1	29.4	(25)
(26)		MCWB	19.9	21.8	32.0	46.0	57.1	63.9	65.7	64.5	56.4	46.8	35.4	27.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	34.8	43.2	56.3	65.1	70.6	71.7	70.6	63.7	53.9	43.0	35.2	(27)
(28)		MDCB	35.5	37.9	49.5	73.5	77.1	83.7	82.7	82.9	76.3	62.7	47.1	39.0	(28)	
(29)		2%	WB	27.6	30.2	37.8	52.2	62.1	68.3	70.4	69.1	61.4	51.5	40.4	32.8	(29)
(30)		MDCB	29.2	32.5	42.1	63.9	75.8	80.4	81.9	81.1	71.8	59.3	43.5	35.2	(30)	
(31)		5%	WB	23.7	25.9	34.8	49.5	60.2	66.4	69.1	67.4	59.4	49.3	38.2	30.7	(31)
(32)		MDCB	25.3	28.8	37.5	59.7	74.1	77.8	79.9	78.3	68.6	55.9	40.7	32.7	(32)	
(33)		10%	WB	20.0	22.1	32.4	46.7	58.2	64.8	67.4	65.6	57.1	47.1	35.7	27.4	(33)
(34)		MDCB	21.6	24.2	34.9	55.7	71.3	76.2	76.9	75.4	63.7	52.8	37.9	29.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.7	20.1	17.4	18.0	20.5	18.3	17.4	16.4	15.8	13.3	10.1	12.6	(35)	
(36)		5% DB	MCDBR	13.5	18.9	17.9	26.6	26.2	23.5	22.8	22.2	21.8	20.5	11.2	8.9	(36)
(37)		MCWBR	12.1	16.2	13.1	13.6	10.7	9.9	8.3	9.0	9.9	11.4	8.5	7.1	(37)	
(38)		5% WB	MCDBR	12.6	17.7	16.2	25.3	24.0	20.7	19.2	20.4	19.4	19.2	11.0	9.0	(38)
(39)		MCWBR	11.6	15.3	12.3	13.6	11.1	10.1	8.5	9.3	10.0	11.4	9.0	7.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.282	0.311	0.367	0.408	0.408	0.405	0.432	0.445	0.385	0.354	0.309	0.280	(40)	
(41)		taud	2.073	2.046	1.978	2.123	2.230	2.310	2.223	2.161	2.293	2.255	2.165	2.048	(41)	
(42)		Ebn at Noon	195	234	246	256	265	267	256	242	243	219	184	163	(42)	
(43)		Edh at Noon	21	31	43	43	41	38	41	41	31	26	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	275	614	1092	1397	1658	1803	1747	1367	929	471	239	171	(44)	
(45)		RadStd	25	34	75	144	160	159	125	135	73	81	27	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page